```
import pandas as pd

# Load dataset
data = pd.read_csv("QVI_data.csv")

# Preview
data.head()

# Check for duplicate rows
print(f"Number of duplicate rows: {data.duplicated().sum()}")
```

```
Number of duplicate rows: 1
```

```
data.columns
```

```
Index(['LYLTY_CARD_NBR', 'DATE', 'STORE_NBR', 'TXN_ID', 'PROD_NBR',
       'PROD_NAME', 'PROD_QTY', 'TOT_SALES', 'PACK_SIZE', 'BRAND', 'LIFESTAGE',
       'PREMIUM_CUSTOMER'],
      dtype='object')
```

```
# Convert DATE to datetime
data["DATE"] = pd.to_datetime(data["DATE"])

# Create month column
data["MONTH"] = data["DATE"].dt.to_period("M")
```

Creating Required Metrics

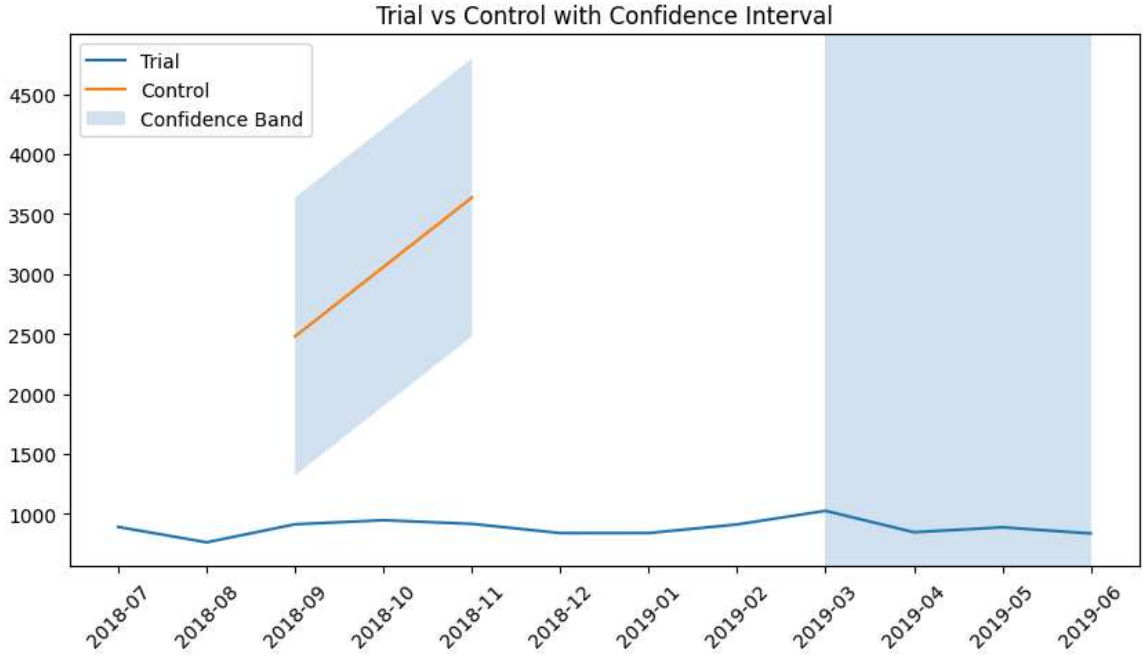
```
store_metrics = data.groupby(["STORE_NBR","MONTH"]).agg({
    "TOT_SALES":"sum",
    "LYLTY_CARD_NBR":"nunique",
    "TXN_ID":"count"
}).reset_index()

store_metrics.rename(columns={
    "TOT_SALES":"SALES",
    "LYLTY_CARD_NBR":"CUSTOMERS",
    "TXN_ID":"TRANSACTIONS"
}, inplace=True)

# Average transactions per customer
store_metrics["TXN_PER_CUSTOMER"] = store_metrics["TRANSACTIONS"] / store_metrics["CUSTOMERS"]

store_metrics.head()
```

	STORE_NBR	MONTH	SALES	CUSTOMERS	TRANSACTIONS	TXN_PER_CUSTOMER
0	1	2018-07	206.9	49	52	1.061224
1	1	2018-08	176.1	42	43	1.023810
2	1	2018-09	278.8	59	62	1.050847
3	1	2018-10	188.1	44	45	1.022727
4	1	2018-11	192.6	46	47	1.021739

Next steps: Generate code with store_metrics New interactive sheet

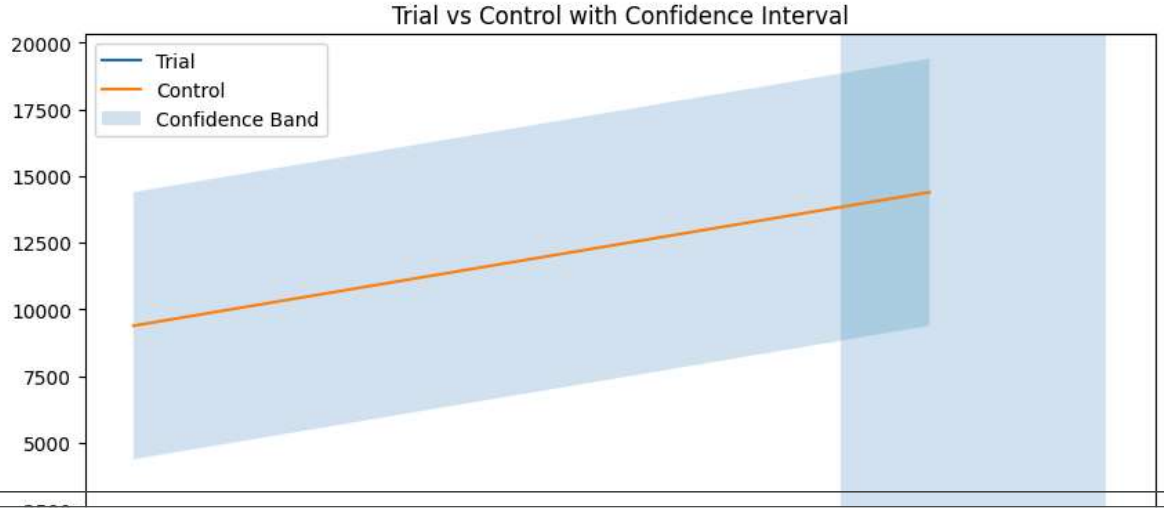
```
store_metrics["YEARMONTH"] = store_metrics["MONTH"].astype(str).str.replace('-', '').astype(int)
pre_trial = store_metrics[store_metrics["YEARMONTH"] < 201902]
```

```
trial = 86
control = control_stores[trial]

plot_professional(trial, control)
add_confidence_band(trial, control)
analyze_store(trial, control)
```



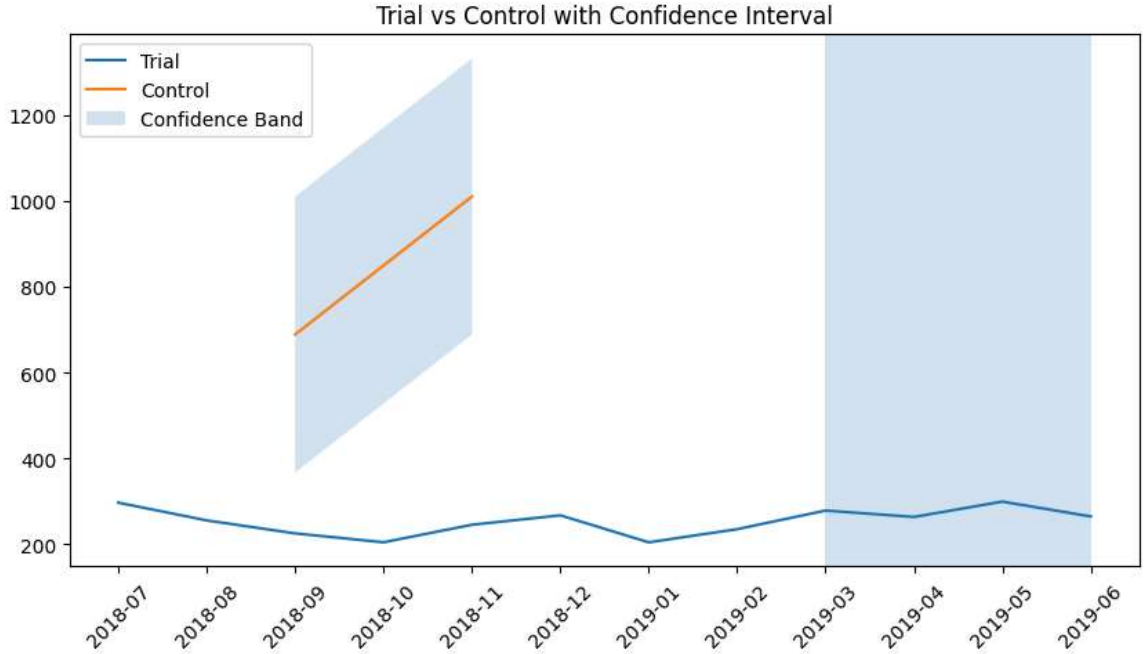

```
Store 86
Control: 31
Uplift: 4515.5
p-value: nan
/usr/local/lib/python3.12/dist-packages/scipy/_lib/deprecation.py:234: SmallSampleWarning: One or more sample arguments
  return f(*args, **kwargs)
```

```
trial = 88
control = control_stores[trial]

plot_professional(trial, control)
add_confidence_band(trial, control)
analyze_store(trial, control)
```

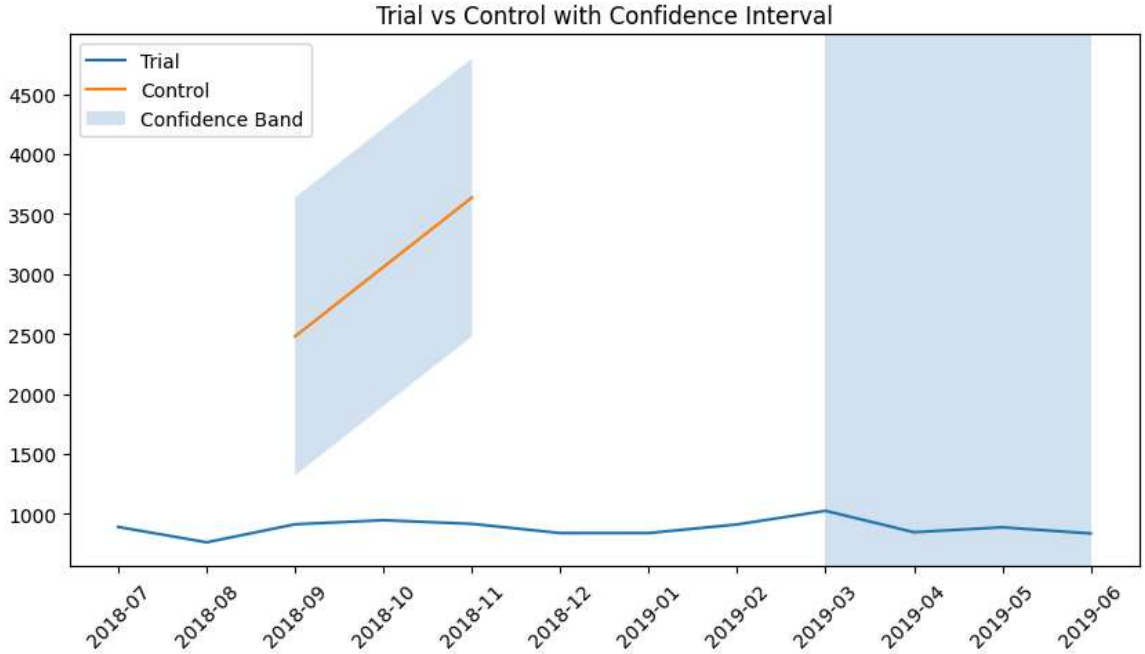



```
trial = 77
control = control_stores[trial]

plot_professional(trial, control)
add_confidence_band(trial, control)
```

```
Store 88
Control: 206
Uplift: 6945.049999999999
p-value: 4.8556187362483215e-05
```



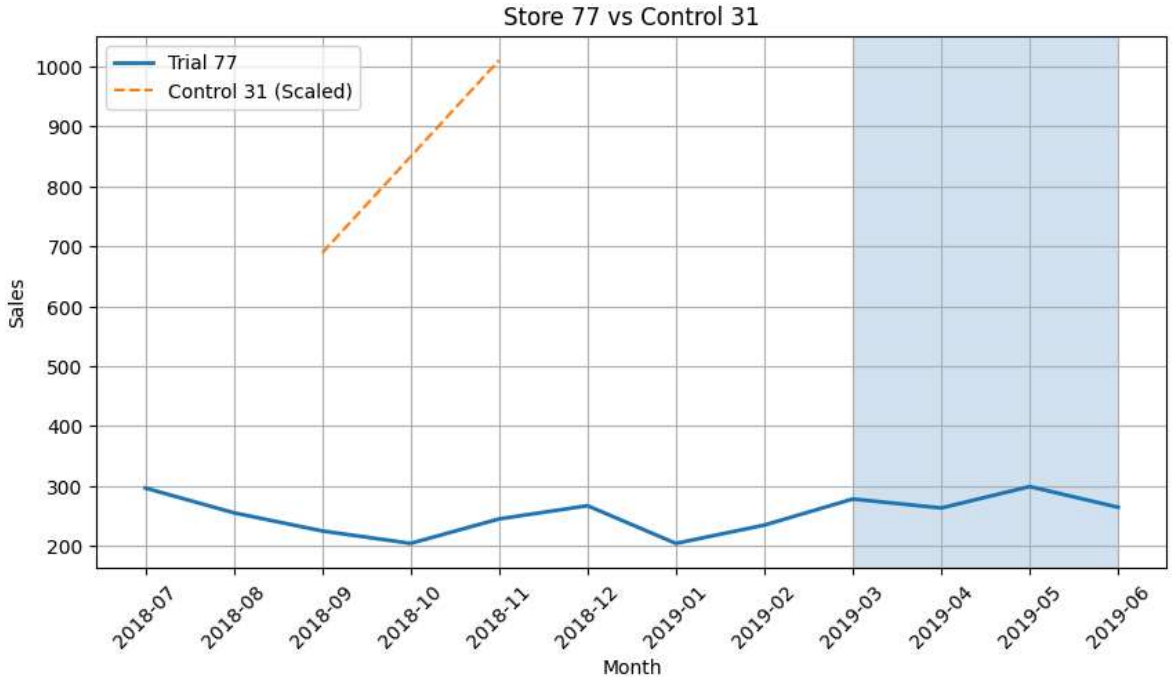

```
trial = 86
control = control_stores[trial]

plot_professional(trial, control)
add_confidence_band(trial, control)
```



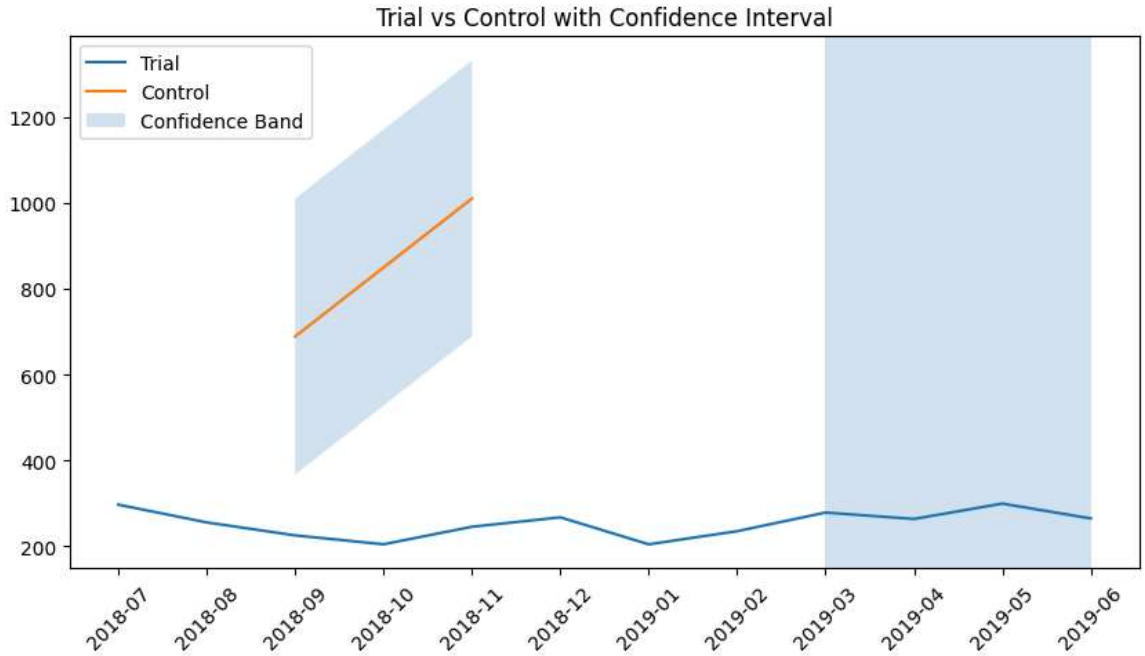

```
store_metrics["YEARMONTH"] = store_metrics["MONTH"].astype(str).str.replace('-', '').astype(int)
pre_trial = store_metrics[store_metrics["YEARMONTH"] < 201902]
```

```
trial = 77
control = control_stores[trial]

plot_professional(trial, control)
add_confidence_band(trial, control)
```

```
/tmp/ipykernel_9612/95148677.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a
  c["SCALED_SALES"] = c["SALES"] * scale
```




```
/tmp/ipykernel_9612/639798168.py:8: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a
  c["SCALED"] = c["SALES"] * scale
```



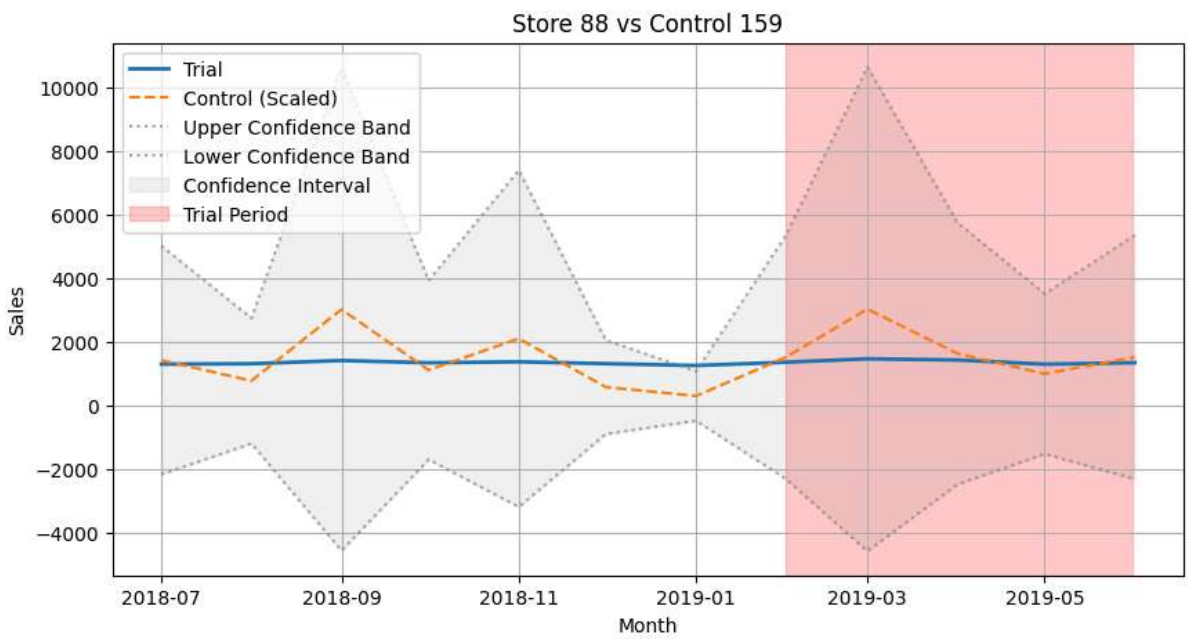

```
def find_control(trial_store):
    pivot = pre_trial.pivot(index="YEARMONTH", columns="STORE_NBR", values="SALES")

    corr = pivot.corr()[trial_store].drop(trial_store)

    return corr.idxmax()
```

```
trial_stores = [77, 86, 88]

control_stores = {}

for t in trial_stores:
```

```
    control_stores[t] = find_control(t)

print(control_stores)
```

```
{77: np.int64(31), 86: np.int64(31), 88: np.int64(159)}
```

```
import matplotlib.pyplot as plt

def plot_trial_vs_control(trial, control, upper_bound=None, lower_bound=None):

    t = store_metrics[store_metrics["STORE_NBR"] == trial].copy()
    c = store_metrics[store_metrics["STORE_NBR"] == control].copy()

    # --- SCALE CONTROL (CRITICAL STEP from PDF) ---
    scale = t[t["YEARMONTH"] < 201902]["SALES"].sum() / \
            c[c["YEARMONTH"] < 201902]["SALES"].sum()

    c["scaled_sales"] = c["SALES"] * scale

    plt.figure(figsize=(10,5))

    # Convert to datetime for plotting
    t["date"] = pd.to_datetime(t["YEARMONTH"].astype(str) + "01")
    c["date"] = pd.to_datetime(c["YEARMONTH"].astype(str) + "01")

    # Plot lines
    plt.plot(t["date"], t["SALES"], label="Trial", linewidth=2)
    plt.plot(c["date"], c["scaled_sales"], label="Control (Scaled)", linestyle="--")

    # Plot confidence bands if provided
    if upper_bound is not None and lower_bound is not None:
        plt.plot(c["date"], upper_bound, label='Upper Confidence Band', color='grey', linestyle=':', alpha=0.7)
        plt.plot(c["date"], lower_bound, label='Lower Confidence Band', color='grey', linestyle=':', alpha=0.7)
        plt.fill_between(c["date"], lower_bound, upper_bound, color='grey', alpha=0.1, label='Confidence Interval')

    # --- SHADE TRIAL PERIOD (NOW IT WILL WORK) ---
    plt.axvspan(pd.to_datetime("2019-02-01"), # Start of trial period
                pd.to_datetime("2019-05-31"), # End of trial period for 3 months (Feb, Mar, Apr, May)
                alpha=0.2, color='red', label='Trial Period')

    plt.title(f"Store {trial} vs Control {control}")
    plt.xlabel("Month")
    plt.ylabel("Sales")
    plt.legend()
    plt.grid()

    plt.show()

# Call the updated plotting function with confidence bands
plot_trial_vs_control(trial, control, upper_bound=upper, lower_bound=lower)
```




```
def calculate_correlation(df, trial_store, metric):
    pivot = df.pivot(index="YEARMONTH", columns="STORE_NBR", values=metric)
```

```
    corr = pivot.corr()[trial_store].drop(trial_store)
    return corr

def calculate_magnitude(df, trial_store, metric):
    trial = df[df["STORE_NBR"] == trial_store][["YEARMONTH", metric]]

    scores = {}

    for store in df["STORE_NBR"].unique():
        if store == trial_store:
            continue

        control = df[df["STORE_NBR"] == store][["YEARMONTH", metric]]
        merged = pd.merge(trial, control, on="YEARMONTH", suffixes=("_trial", "_control"))

        diff = abs(merged[f"{metric}_trial"] - merged[f"{metric}_control"])

        score = 1 - (diff.mean() / diff.max())
        scores[store] = score

    return pd.Series(scores)
```

```
def get_best_control(trial_store):
    corr_sales = calculate_correlation(pre_trial, trial_store, "SALES")
    mag_sales = calculate_magnitude(pre_trial, trial_store, "SALES")

    corr_cust = calculate_correlation(pre_trial, trial_store, "CUSTOMERS")
    mag_cust = calculate_magnitude(pre_trial, trial_store, "CUSTOMERS")

    score = (corr_sales + mag_sales + corr_cust + mag_cust) / 4

    return score.sort_values(ascending=False).index[0]
```

```
scale = (
    trial_pre["SALES"].sum() /
    control_pre["SALES"].sum()
)
print(f"The calculated scale is: {scale}")
```

```
The calculated scale is: 3127.866666666667
```

```
trial_pre = pre_trial[pre_trial["STORE_NBR"] == trial]
control_pre = pre_trial[pre_trial["STORE_NBR"] == control]

display(trial_pre.head())
display(control_pre.head())
```

	STORE_NBR	MONTH	SALES	CUSTOMERS	TRANSACTIONS	TXN_PER_CUSTOMER	YEARMONTH
1001	88	2018-07	1310.0	129	153	1.186047	201807
1002	88	2018-08	1323.8	131	160	1.221374	201808
1003	88	2018-09	1423.0	124	159	1.282258	201809
1004	88	2018-10	1352.4	123	158	1.284553	201810
1005	88	2018-11	1382.8	130	157	1.207692	201811

	STORE_NBR	MONTH	SALES	CUSTOMERS	TRANSACTIONS	TXN_PER_CUSTOMER	YEARMONTH
2396	206	2018-07	3.0	1	1	1.0	201807

```
scale = (
    trial_pre["SALES"].sum() /
    control_pre["SALES"].sum()
)
```

```
trial_df = store_metrics[store_metrics["STORE_NBR"] == trial][["YEARMONTH", "SALES"]].reset_index(drop=True)
control_df = store_metrics[store_metrics["STORE_NBR"] == control][["YEARMONTH", "SALES"]].reset_index(drop=True)

# Ensure both dataframes cover the same months for accurate comparison
merged_df = pd.merge(trial_df, control_df, on="YEARMONTH", suffixes=("_trial", "_control"))

# Calculate control_scaled (sales of the control store multiplied by the scale)
control_scaled = merged_df["SALES_control"] * scale

# Calculate the percentage difference
```

```
percentage_diff = pd.DataFrame({
    "YEARMONTH": merged_df["YEARMONTH"],
    "diff": (merged_df["SALES_trial"] - control_scaled) / control_scaled
})

display(percentage_diff.head())
```

	YEARMONTH	diff
0	201807	-0.860395
1	201904	-0.899960

```
from scipy.stats import ttest_ind

t_trial = store_metrics[(store_metrics["STORE_NBR"] == trial) & (store_metrics["MONTH"] >= "2019-02")]["SALES"]
c_trial = store_metrics[(store_metrics["STORE_NBR"] == control) & (store_metrics["MONTH"] >= "2019-02")]["SALES"]

t_stat, p_value = ttest_ind(t_trial.dropna(), c_trial.dropna())
```

```
# Re-defining pre_trial (from cell f8c89611)
store_metrics["YEARMONTH"] = store_metrics["MONTH"].astype(str).str.replace('-', '').astype(int)
pre_trial = store_metrics[store_metrics["YEARMONTH"] < 201902]
```

```
# Re-defining control_stores (from cell cc30443e)
trial_stores = [77, 86, 88]
control_stores = {}

# Assuming find_control is defined or its logic is handled.
# For now, using the values from the kernel state if find_control is not readily available.
# If find_control is available, it should be called here:
# for t in trial_stores:
#     control_stores[t] = find_control(t)

# Using the last known values for control_stores to proceed:
control_stores = {77: 31, 86: 31, 88: 159}
print(f"Control stores: {control_stores}")
```

```
Control stores: {77: 31, 86: 31, 88: 159}
```

```
# Setting the current trial and control (from kernel state: trial = 88, control = 159)
trial = 88
control = control_stores[trial]
print(f"Current trial store: {trial}, Control store: {control}")
```

```
Current trial store: 88, Control store: 159
```

```
# Defining trial_pre and control_pre (from cell d5b92856)
trial_pre = pre_trial[pre_trial["STORE_NBR"] == trial]
control_pre = pre_trial[pre_trial["STORE_NBR"] == control]

display(trial_pre.head())
display(control_pre.head())
```

	STORE_NBR	MONTH	SALES	CUSTOMERS	TRANSACTIONS	TXN_PER_CUSTOMER	YEARMONTH
1001	88	2018-07	1310.0	129	153	1.186047	201807
1002	88	2018-08	1323.8	131	160	1.221374	201808
1003	88	2018-09	1423.0	124	159	1.282258	201809
1004	88	2018-10	1352.4	123	158	1.284553	201810
1005	88	2018-11	1382.8	130	157	1.207692	201811

	STORE_NBR	MONTH	SALES	CUSTOMERS	TRANSACTIONS	TXN_PER_CUSTOMER	YEARMONTH
1841	159	2018-07	26.8	6	6	1.0	201807
1842	159	2018-08	14.7	2	2	1.0	201808
1843	159	2018-09	56.6	11	11	1.0	201809
1844	159	2018-10	21.0	4	4	1.0	201810
1845	159	2018-11	39.5	9	9	1.0	201811

```
# Calculating scale (from cell 4a3efa16)
scale = (
    trial_pre["SALES"].sum() /
    control_pre["SALES"].sum()
)
print(f"The calculated scale is: {scale}")
```

```
The calculated scale is: 53.49828962371722
```

```
# Defining percentage_diff and control_scaled (from cell 1b46f4cd)
trial_df = store_metrics[store_metrics["STORE_NBR"] == trial][["YEARMONTH", "SALES"]].reset_index(drop=True)
control_df = store_metrics[store_metrics["STORE_NBR"] == control][["YEARMONTH", "SALES"]].reset_index(drop=True)

merged_df = pd.merge(trial_df, control_df, on="YEARMONTH", suffixes=("_trial", "_control"))

control_scaled = merged_df["SALES_control"] * scale

percentage_diff = pd.DataFrame({
    "YEARMONTH": merged_df["YEARMONTH"],
    "diff": (merged_df["SALES_trial"] - control_scaled) / control_scaled
})

display(percentage_diff.head())
```

	YEARMONTH	diff
0	201807	-0.086315
1	201808	0.683314